Tribhuvan University

Faculty of Humanities and Social Science

A PROJECT REPORT ON

SPORTS EQUIPMENT RENTAL SYSTEM

Submitted to

Department of Computer Application

Saraswati Multiple Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by DIWAS LAMICHHANE & SAILESH SHRESTHA entitled **"SPORTS EQUIPMENT RENTAL SYSTEM"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

……

Er. Manoj Giri

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LETTER OF APPROVAL

This is to certify that this project prepared by **DIWAS LAMICHHANE & SAILESH SHRESTHA** entitled **"SPORTS EQUIPMENT RENTAL SYSTEM"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application (BCA). In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

Signature of Supervisor …… **Er. Manoj Giri** Saraswati Multiple Campus	Signature of HOD …… **Mrs. Nanu Devkota**, BCA, HOD Saraswati Multiple Campus
Signature of Internal Examiner ……..…….. **Internal Examiner**	Signature of External Examiner …… **External Examiner**

ABSTRACT

The Sports Equipment Rental System is a web-based application developed to automate and manage the process of renting sports equipment efficiently. The system eliminates manual record-keeping and provides a digital platform for users to browse, request, and rent equipment, while enabling administrators to manage inventory and user requests effectively.

The system is built using HTML, CSS, JavaScript, and Bootstrap for the frontend, and PHP with MySQL for backend processing and database management. It provides two main interfaces: a User Panel and an Admin Panel. In the User Panel, users can register, log in, and view available sports equipment. They can send rental requests and track their request status. The Admin Panel allows administrators to manage equipment inventory, approve or reject rental requests, and maintain records of issued and returned items through CRUD operations.

A major feature of this system is the integration of a Recommendation Algorithm, which suggests suitable sports equipment to users based on factors such as previous rentals, user preferences, and equipment availability. This improves user experience by helping users quickly find relevant equipment without manually searching through the entire inventory.

Keywords: ***Sports Equipment Rental System, Digital Platform, User Panel, Admin Panel, Recommendation Algorithm, User-friendly, Efficient***

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Diwas Lamichhane

Sailesh Shrestha

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LIST OF ABBREVIATIONS

Abbreviation	Definition
BS	Bootstrap
CRUD	Create, Read, Update and Delete
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ER	Entity Relationship
HTML	Hypertext Markup Language
IDE	Integrated Development Environment
JS	Java Script
MySQL	Microsoft Server Structured Query Language
PHP	Hypertext Preprocessor
SDLC	Software Development Lifecycle
SERS	Sports Equipment Rental System
UI	User Interface
UX	User Experience

CHAPTER 1: INTRODUCTION

1.1 Introduction

Sports Equipment Rental System is a web-based application designed to simplify and manage the process of renting sports equipment efficiently. The system provides a centralized platform where users can browse available equipment, request items, and manage their rentals, while administrators can monitor and control all activities from a single dashboard.

This system enhances convenience for users by allowing them to search for equipment, view details, sort items in ascending order, and request equipment online. It also includes a recommendation feature that suggests suitable equipment based on user interaction. Users can create accounts, log in to the system, manage their profiles, and provide feedback.

The project is developed using HTML, CSS, JavaScript, and Bootstrap for the frontend, while PHP is used as the backend for handling server-side operations. MySQL is used for database management, and XAMPP serves as the local development environment. The system consists of two main modules: Admin Module and User Module.

Admin Module:

The Admin Panel allows administrators to manage equipment inventory, approve or reject rental requests, and maintain records of issued and returned items through CRUD operations.

- **Equipment Management:**
  Allows the administrator to add, view, and delete equipment records, including equipment name, category, and availability.
- **User Information:**
  Enables the admin to view registered user details such as id, first name, last anme, username, email, and mobile number, excluding sensitive data like passwords.
- **Issue Information:**
  Displays details of borrowed equipment, including user information, issue date, return date, and current status (issued, returned, or expired).
- **Expired List:**
  Shows the list of equipment that has or has not been returned within the due date which is marked as returned or expired.

User Module:

In the User Panel, users can register, log in, and view available sports equipment. They can send rental requests and track their request status.

- **Home Page:**
  Provides an overview of the system and highlights available services.
- **Equipment Page:**
  Displays all available sports equipment with details such as id, name, category, quantity, rental price per day, condition and status.
- **Search and Sorting:**
  Users can search for specific equipment and sort the list in ascending order for better usability.
- **Equipment Request:**
  Allows users to request equipment for rental.
- **Recommendation System:**
  Suggests equipment to users based on their popularity and times the item has been requested.
- **Profile:**
  Users can view and manage their personal information.
- **Feedback:**
  Users can submit feedback or queries to the system administrators.
- **Fines:**
  Users can see their fines list given the date assigned has been expired
- **Login/Logout:**
  Provides secure authentication for users to access system features.
- **Side Navigation Features:**
  - **Equipment Request List:** View requested items
  - **Issue Information:** View currently borrowed equipment
  - **Expired List:** View overdue equipment

1.2 Problem Statement

Traditional sports equipment rental systems are often managed manually, leading to several issues such as:

- Inefficient record-keeping and data management
- Difficulty in tracking equipment availability
- High chances of human errors and data inconsistency
- Time-consuming rental and approval processes
- Lack of personalized recommendations for users

Users often struggle to find suitable equipment quickly, while administrators face challenges in managing inventory and requests effectively. Therefore, there is a need for an automated system that simplifies the rental process and improves overall efficiency.

1.3 Objective

- To develop a web-based system that enables users to browse, request, and track sports equipment rentals with secure authentication.
- To automate equipment management, including inventory tracking, rental processing, and administrative control through CRUD operations.
- To enhance user experience by implementing a recommendation algorithm and ensuring accurate availability tracking to prevent double booking.

1.4 Scope and Limitation

Scope

- The system is accessible through a web browser
- Users can register, log in, and request sports equipment
- Admin can manage equipment, users, and rental approvals
- Real-time tracking of equipment availability
- Recommendation system for better user experience

Limitations

- The system is limited to a single organization or rental provider
- Online payment integration is not included (if not implemented)
- Recommendation algorithm is basic and based on limited factors
- Requires internet connection to access the system
- Security features are limited to basic authentication mechanisms

1.5 Development Methodology

The development of the Sports Equipment Rental System will follow the Waterfall Model, a linear and sequential development approach. In this methodology, each phase of the development process is completed before moving on to the next, ensuring a structured and well-documented system design.

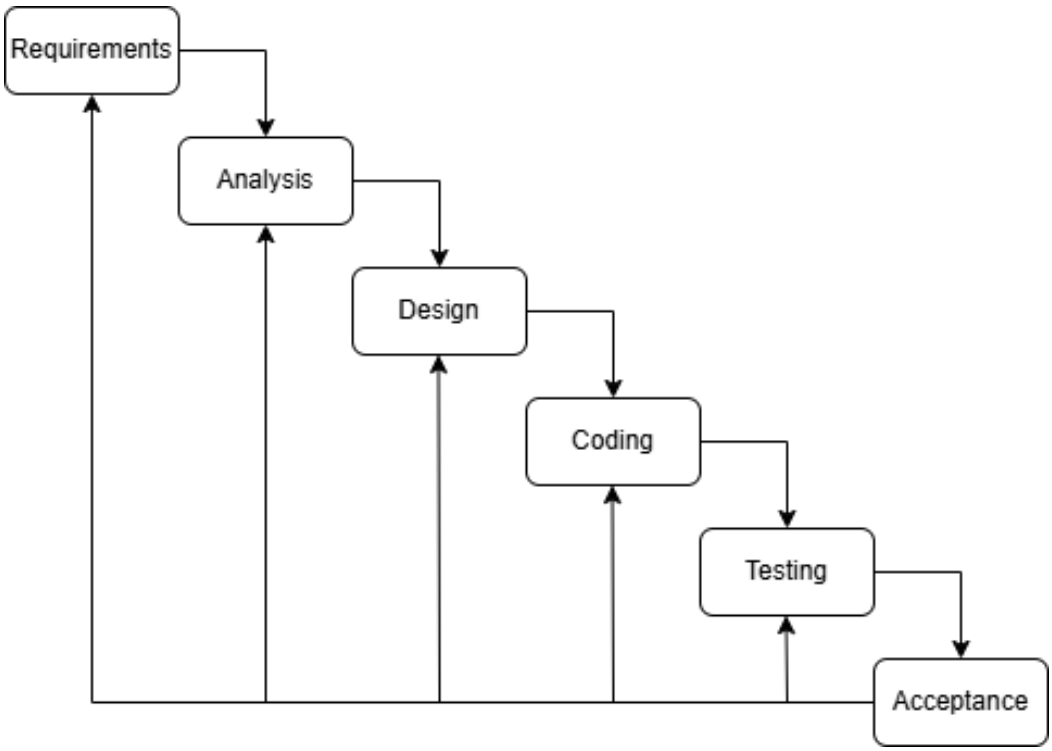

Figure 1 Iterative Waterfall Model for SERS

The development process begins with the requirement gathering phase, where all system functionalities such as equipment management, user authentication, request handling, and admin controls are identified. This is followed by the analysis phase, where the feasibility and system requirements are studied in detail.

In the design phase, the system architecture, database structure, and user interface layouts are planned. After that, the coding phase is carried out using technologies such as HTML, CSS, JavaScript, PHP, and MySQL to implement the system functionalities.

Once implementation is completed, the system enters the testing phase, where errors and bugs are identified and corrected. During this phase, necessary modifications are made by revisiting earlier stages, which reflects the iterative nature of the model. Finally, in the

acceptance phase, the system is verified to ensure it meets all user requirements and functions correctly.

Thus, the Iterative Waterfall Model provides a structured yet flexible approach, ensuring systematic development along with continuous improvement of the system.

1.6 Report Organization

This report document contains five chapters they are as follows:

Chapter 1: Introduction

This chapter provides an overview of the system, including the introduction, problem statement, objectives, scope, and limitations.

Chapter 2: Background Study and Literature Review

This chapter presents the background study and analysis of existing fitness applications and related research.

Chapter 3: System Analysis and Design

This chapter includes requirement analysis, system architecture, algorithm used and design methodologies.

Chapter 4: Implementation and Testing

This chapter describe the implementation process, testing methods, and bug handling techniques.

Chapter 5: Conclusion and Further Enhancement

This chapter summarizes the project, discusses limitations, and provides recommendations for future improvements.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Sports participation is a fundamental aspect of physical well-being and recreational activity worldwide, with numerous clubs, universities, and communities providing access to various sports equipment. Traditionally, managing the inventory, issuance, and return of this equipment has been done manually, leading to inefficiencies in tracking availability, monitoring return deadlines, and managing administrative records. The need for a digital system to streamline operations has grown with the increasing volume of equipment and the demand for real-time accessibility.

A Sports Equipment Rental System (SERS) provides an integrated solution for handling equipment inventory, processing user requests, and managing rental information efficiently. Web-based management systems have proven to be more effective in ensuring data consistency, minimizing redundancy, and enhancing accessibility compared to traditional paper-based or spreadsheet-based systems.

Implementing a web-based management system enhances operational efficiency, provides centralized data storage, and improves user accessibility. Using PHP and MySQL, the system ensures real-time data updates and secure handling of inventory and user records. Moreover, by utilizing Vanilla CSS, HTML, and JavaScript, the system offers an intuitive and responsive user interface, making it accessible on multiple devices. Additionally, the integration of a frequency-based recommendation system helps users quickly identify popular equipment, further enhancing the user experience.

2.2 Literature Review

The development of a Sports Equipment Rental System (SERS) using PHP and MySQL has become increasingly significant due to the growing need for automated inventory management in sports facilities. The system is designed to manage various aspects such as equipment registration, request approval, and deadline tracking, which are typically handled manually in smaller organizations. The growing complexity of managing large volumes of sports gear calls for integrated solutions that can enhance efficiency and accuracy in managing rental transactions.

In recent years, various web-based systems have been proposed for managing resource rentals. For instance, an advanced system for sports equipment management was developed to handle real-time availability and user assignments. This system employs PHP and MySQL to manage backend operations while providing an interactive user interface for both admins and members [1].

The integration of PHP and MySQL in equipment management systems is emphasized in the work by Sharma, where a robust PHP-based management system for resource tracking is demonstrated. The system allows admin users to perform CRUD (Create, Read, Update, Delete) operations for inventory management, making the process efficient and reducing human error. Additionally, the system provides an interactive front-end for users to browse and request equipment based on their needs [2].

A broader perspective on resource management systems, including web-based solutions for various rental domains, has been discussed in several research papers. For example, Smith et al. highlight the importance of web-based systems in the management of institutional assets, stressing the benefits of data centralization, which allows administrators and users to access crucial information anytime, anywhere. These systems typically include features like status tracking, automated fine calculations, and history logs, which are essential for improving operational efficiency [3].

Moreover, recent advancements in software engineering have been geared toward optimizing the user experience through recommendation algorithms. A web-based system for managing rentals is no exception. The literature in this area underlines the need for responsive design, efficient database management, and intelligent data aggregation to surface popular items to users [4].

Finally, several educational resources and tutorials on creating inventory management systems provide practical insights and code examples that help developers implement these systems. One such resource by Project Worlds offers a step-by-step guide for building a rental management system using PHP and MySQL, demonstrating the power of these technologies in developing scalable and secure web applications for resource management [5].

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

System analysis is the process of studying a procedure or business to identify its goals and purposes and create systems and procedures that will achieve them in an efficient way. For the Sports Equipment Rental System (SERS), a structured approach has been adopted to define the system's functional and non-functional requirements.

3.1.1 Requirement Analysis

Requirement analysis is a critical phase that determines the needs or conditions to meet for a new or altered product.

i. **Functional Requirements:**

Functional requirements describe what the system should do. These are illustrated through the interactions between the two primary actors: User and Admin.

User Module:

In the User Panel, users can register, log in, and view available sports equipment. They can send rental requests and track their request status.

- **Registration & Authentication:** Users must be able to register and log in securely.
- **Equipment Catalog:** Users can browse all available sports equipment with details like category.
- **Search Facility:** Users can search for specific equipment by name.
- **Borrow Request:** Users can submit a request to borrow an item for a specific duration.
- **Issued Equipment Tracking:** Users can view the status of their requests and details of currently borrowed items.
- **Recommendation System:** Users can view "Recommended Equipments" based on the most frequently requested items (popularity based).

Admin Module:

- **Inventory Management:** Admin has full control to add new equipment, update existing details, or delete items.
- **Request Processing:** Admin can view pending requests and choose to approve or deny them.

- **Issue Tracking:** Admin can monitor all issued equipment and associated return dates.
- **Expiration Management:** A dedicated view for "Expired List" to track equipment not returned by the deadline.
- **Feedback Review:** Admin can view feedback submitted by users to improve service.

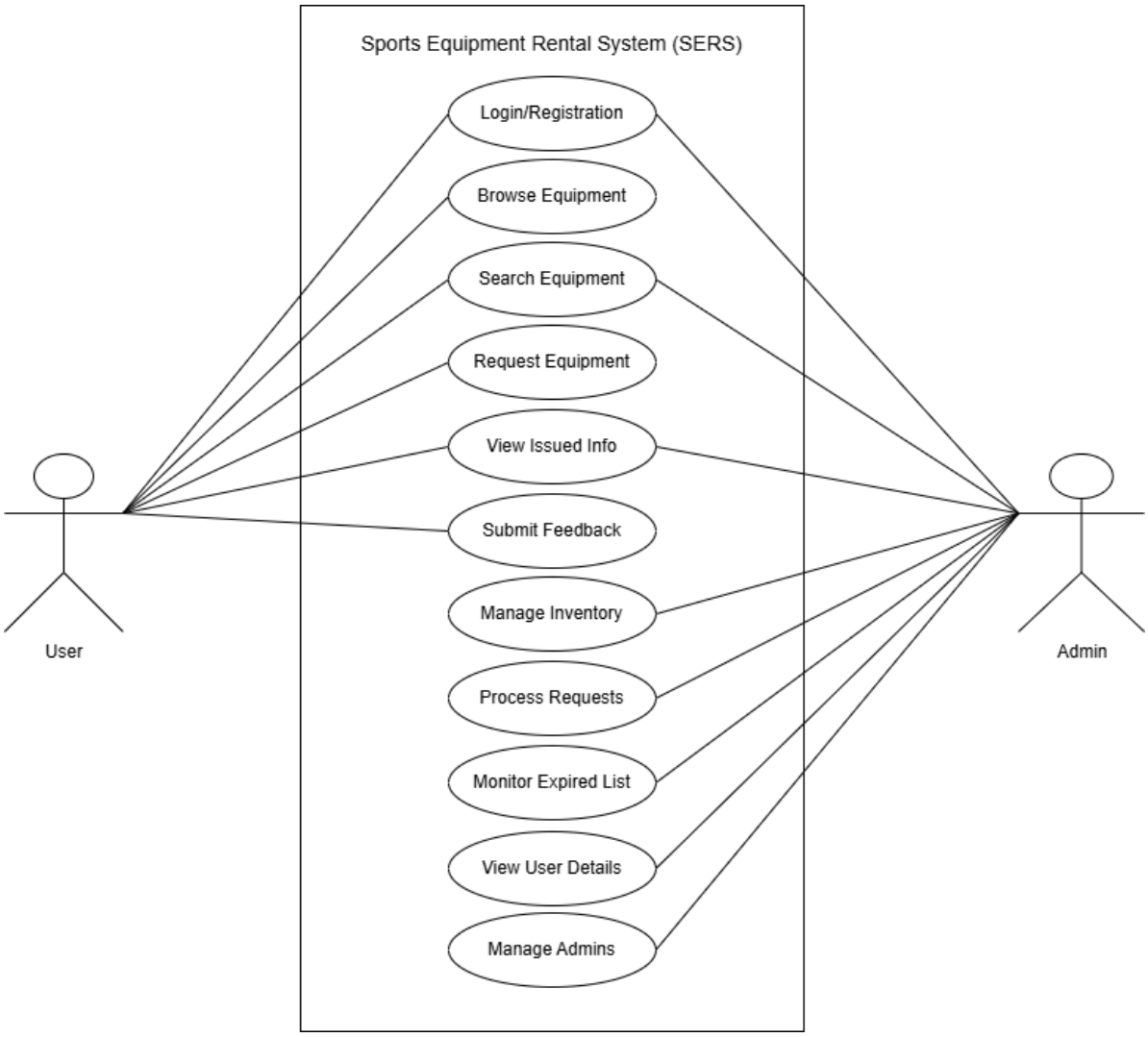

Figure 2 Use Case Diagram for SERS

ii. Non-Functional Requirements:

- **Usability:** The system uses Vanilla CSS for a clean, intuitive, and responsive interface compatible with various screen sizes.
- **Security:** Unauthorized access is prevented through session-based authentication for both user and admin panels.

- **Performance:** The system ensures low latency in database queries using optimized SQL and PHP backend processing.
- **Reliability:** Database constraints in MySQL ensure that data (like equipment quantities) remains consistent during transactions.

3.1.2 Feasibility Analysis

A feasibility study is an assessment of the practicality of a proposed project.

1. Technical Feasibility:

The project utilizes the XAMPP stack (Apache, MySQL, PHP), which is highly stable and widely documented. The hardware requirements are minimal, as it can run on any standard PC acting as a local server.

2. Operational Feasibility:

The system is designed with a simple workflow that mirrors real-world rental processes, making it easy for sports department staff and students to adopt without extensive training.

3. Economic Feasibility:

Being built with open-source technologies, there are no software licensing costs. The long-term savings from reduced manual paperwork and better equipment tracking justify the development effort.

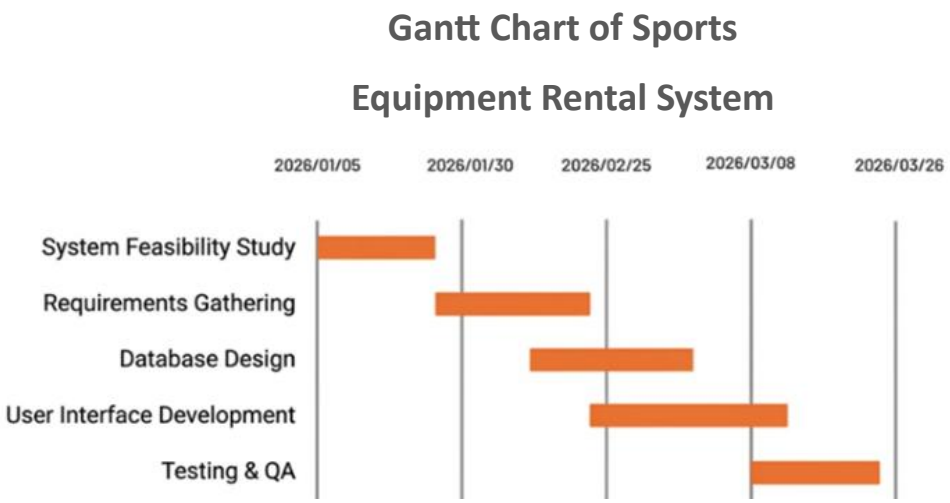

Figure 3 Gantt Chart for SERS

3.1.3 Data Modeling

Data modelling involves defining the data structures and relationships. For SERS, the Entity-Relationship (ER) logic is as follows:

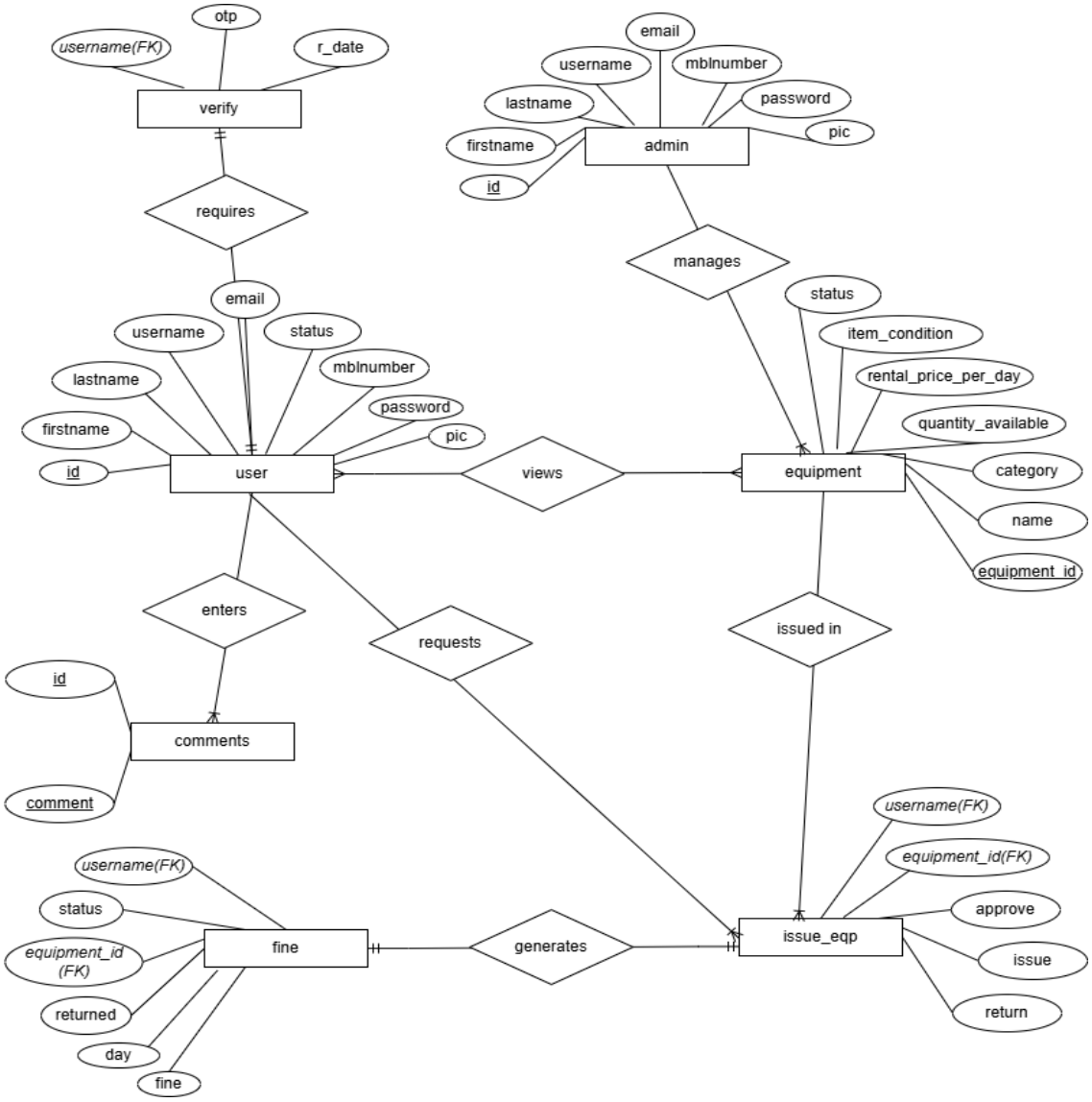

Figure 4 Entity-Relationship Diagram for SERS

3.1.4 Process Modeling

Data Flow Diagrams show the flow of data from external entities into the system, and from one process to another within the system. Following are the Data Flow Diagrams for the current system. Each process within the system is first shown as a Context Level DFD and later as a Detailed DFD. The Context Level DFD provides a conceptual view of the process and its surrounding input, output and data stores. The Detailed DFD provides a more

detailed and comprehensive view of the interaction among the sub processes within the system which is explained below in the figure:

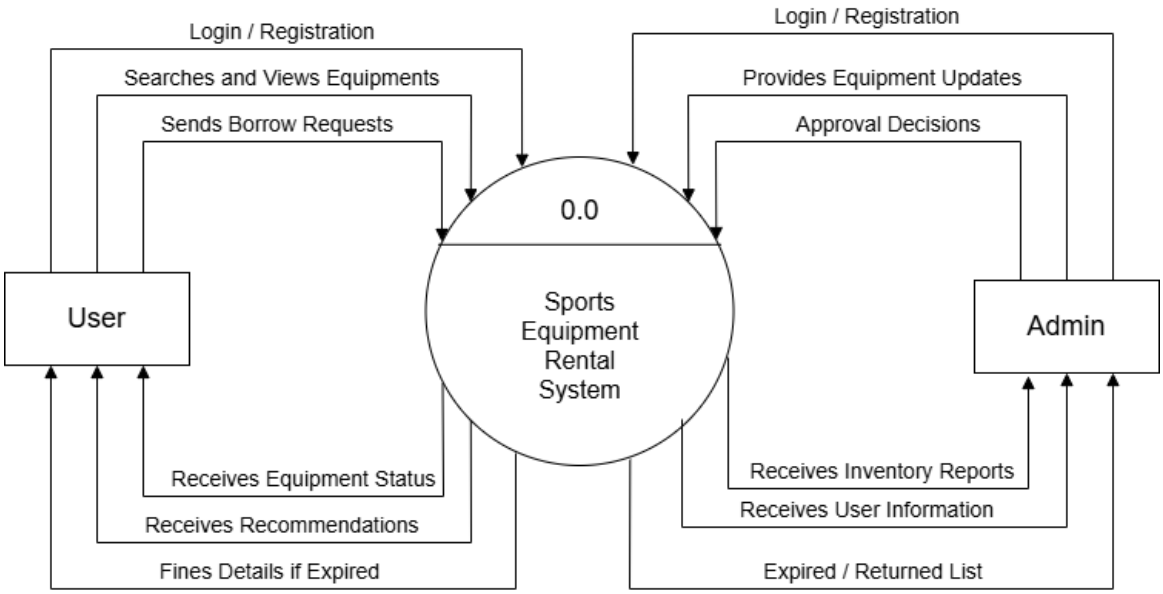

Figure 5 Level 0 Data Flow Diagram for SERS

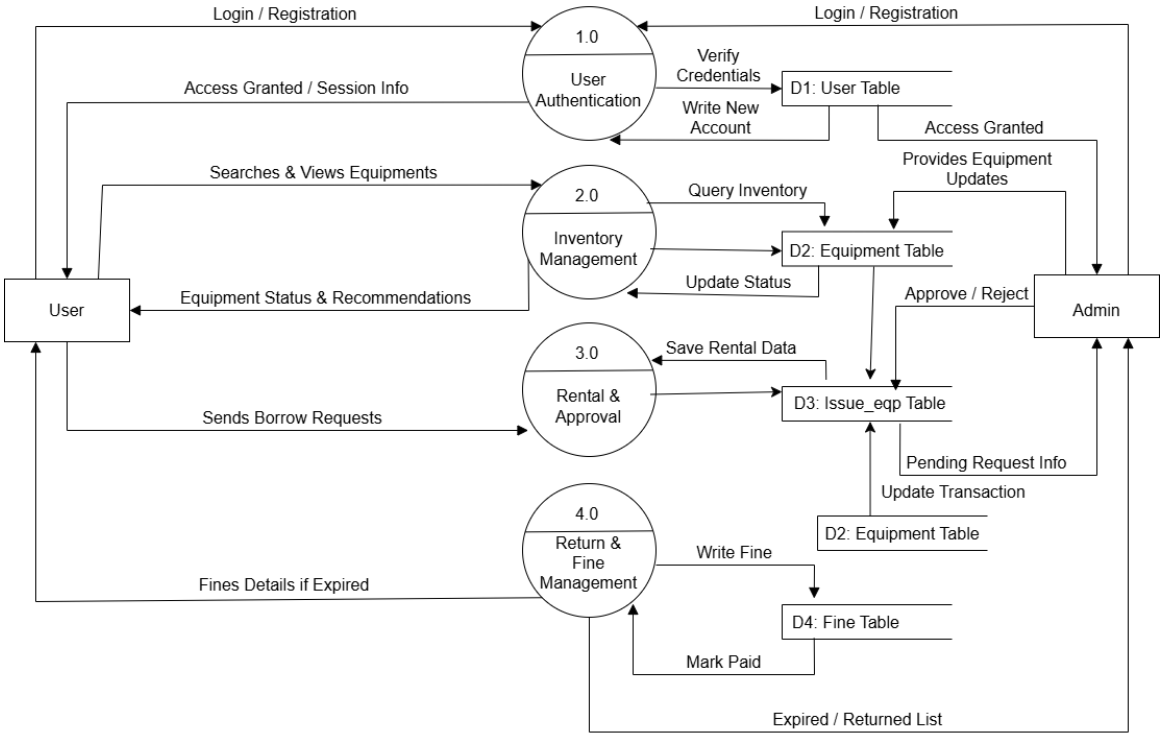

Figure 6 Level 1 Data Flow Diagram for SERS

3.2 System Design

To effectively illustrate the diverse functional requirements of the system, various design diagrams have been created. These diagrams serve to visually represent the system's architecture and functionalities, ensuring clarity and understanding among stakeholders.

3.2.1 Architecture Design

The SERS is built using a 3-Tier Architecture, which separates the application into three logical and physical computing tiers:

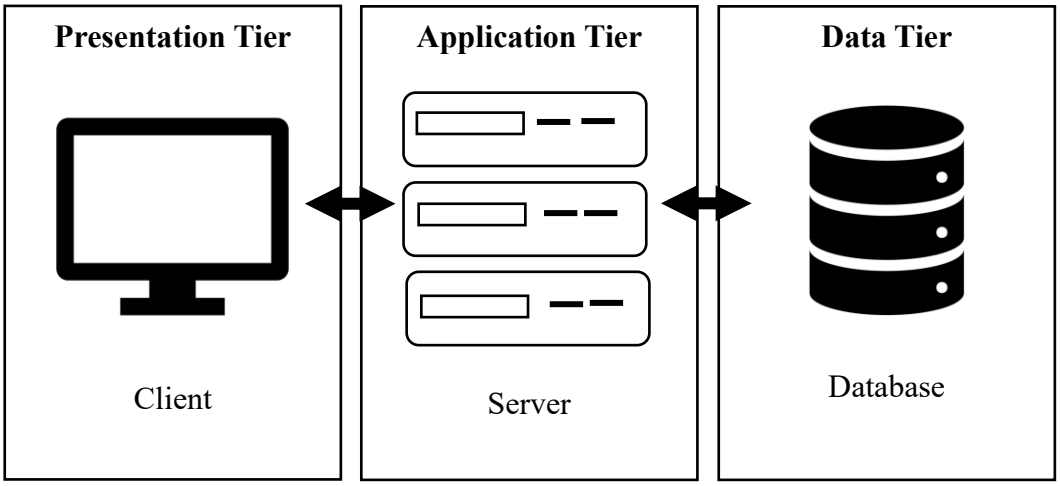

Figure 7 Architectural Design for SERS

- **Presentation Tier (Client Layer):**

This is the topmost level of the application. It consists of the user interface built with HTML5, CSS, and JavaScript. It communicates with the application tier through HTTP requests.

- **Application Tier (Logic Layer):**

This tier is developed using PHP. It handles the core functional logic of the system, such as processing rental requests, calculating fines, and running the recommendation algorithm. It acts as a bridge between the user interface and the database.

- **Data Tier (Database Layer):**

This tier consists of the MySQL database server. It stores all persistent data, including user profiles, equipment inventory, transaction logs, and feedback.

3.2.2 Database Schema Design

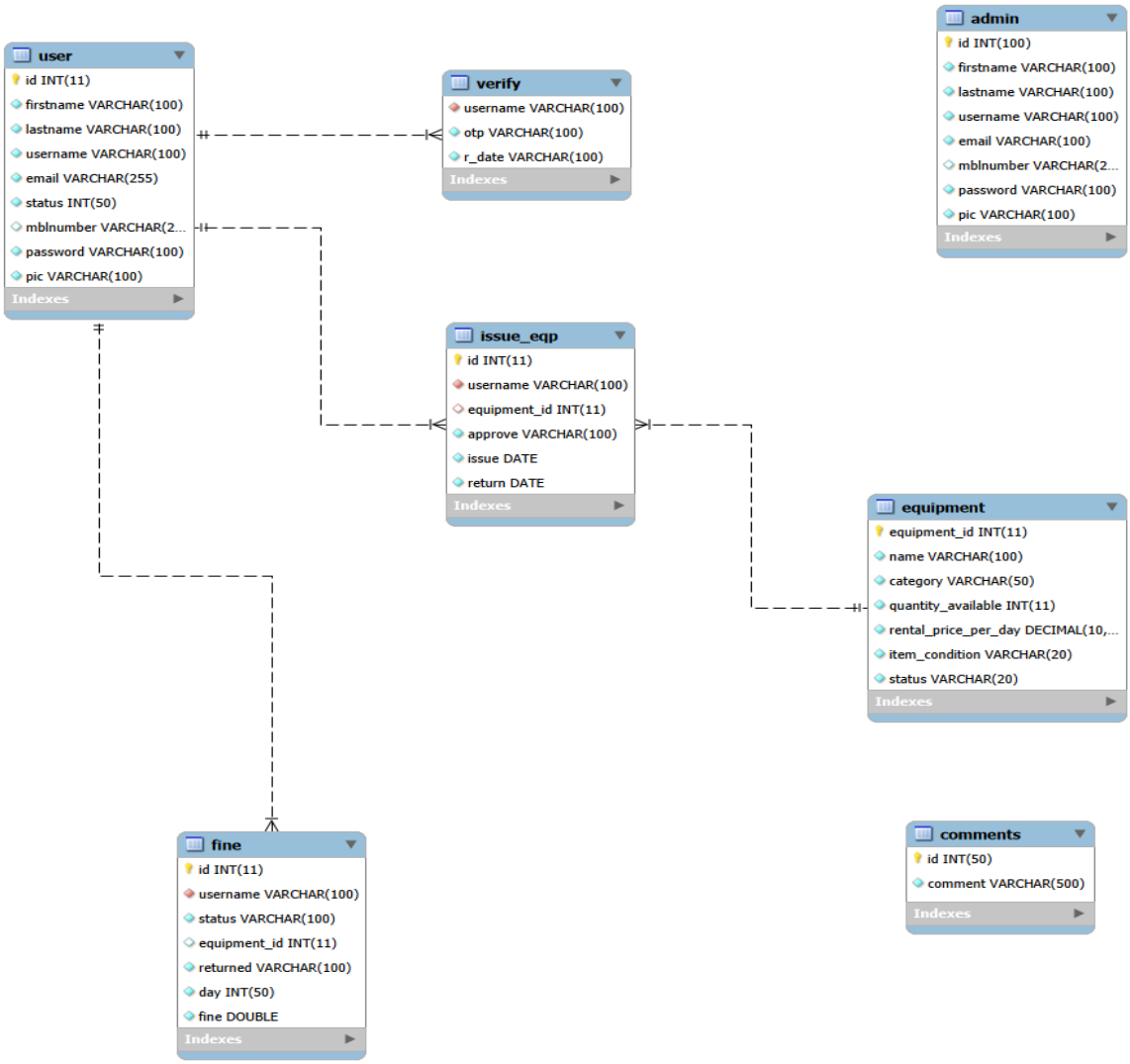

Figure 8 Database Schema Design for SERS

3.2.3 Interface Design

The user interface of the application has been designed using Figma. It is an open-source free web-based UI tool to create, collaborate, prototype, and handoff. The user interface of the system is designed to be simple, clear, and user-friendly. It provides an easy navigation structure with a navigation bar, sidebar, and main content area. Users can quickly access different features such as viewing equipment, making requests, and managing data. The design focuses on clarity and ease of use so that even new users can interact with the system without confusion. Proper alignment, spacing, and consistent layout are maintained throughout the system to enhance the overall user experience.

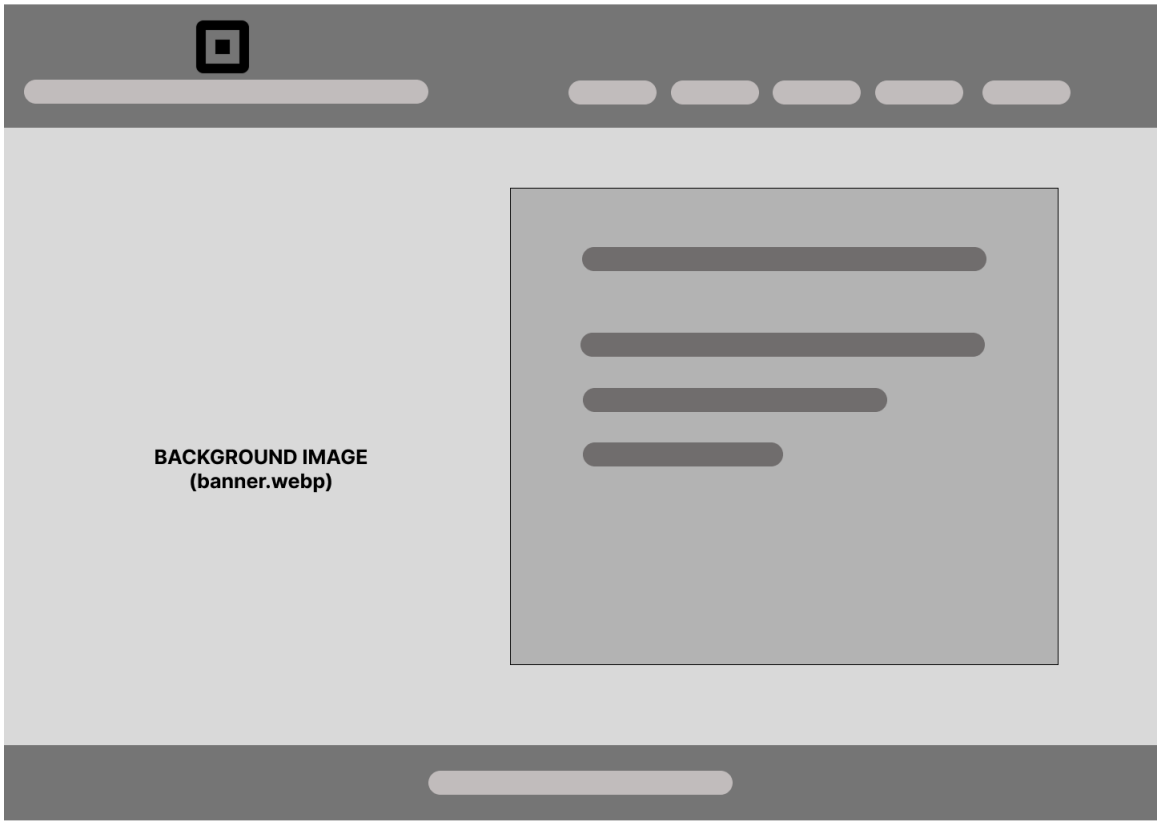

Figure 9 User Interface Design for Home Page

Figure 10 User Interface Design for Login Page

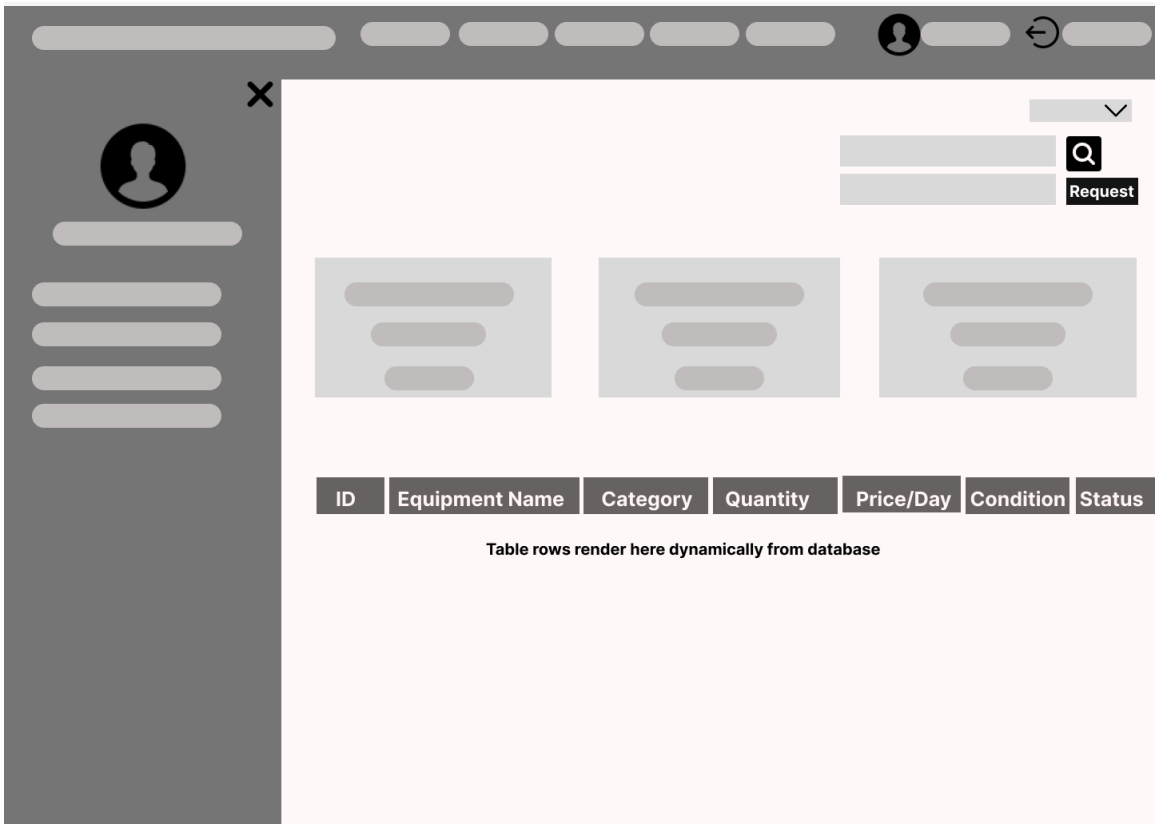

Figure 11 User Interface Design for Equipments Page for User

3.2.4 Physical DFD

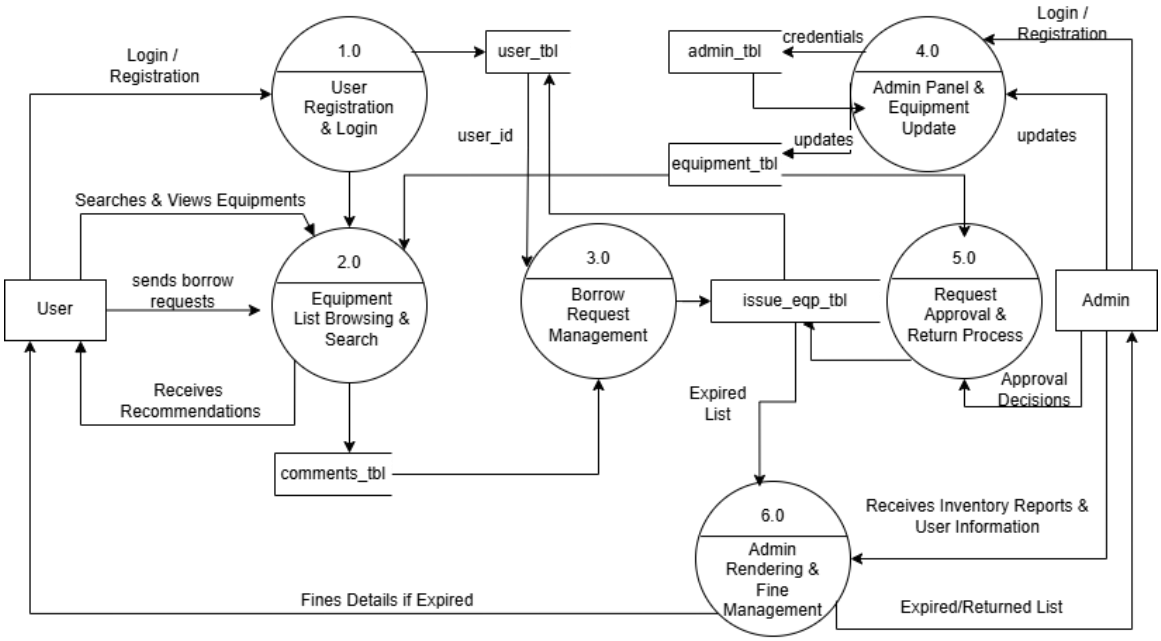

Figure 12 Physical Data Flow Diagram for SERS

3.3 Algorithm Details

The SERS system utilizes specific algorithms to automate administrative tasks and enhance the user experience. The two primary algorithms are detailed below:

3.3.1 Fine Calculation Algorithm

This algorithm is used to calculate penalties for equipment that is not returned by the due date. It ensures that users are held accountable for late returns.

Algorithm Steps:

1. **Input:** The `return_date` from the `issue_book` table and the `current_date` from the system.
2. **Comparison:** Check if `current_date > return_date`.
3. **Duration Calculation:** If true, calculate the difference between the two dates in seconds using `strtotime()`.
4. **Conversion:** Convert the difference from seconds to days: `Days = seconds / (60 * 60 * 24) `.
5. **Penalty Calculation:** Multiply the number of late days by a fixed rate (e.g., Rs. 10 per day): `Fine = Days * 10`.
6. **Action:** Update the status of the transaction to "EXPIRED" and log the fine details.

3.3.2 Recommendation Algorithm

The recommendation system uses a popularity-based approach (most borrowed) to suggest equipment to users. This helps users discover items that are frequently used by others.

Algorithm Steps:

1. Data Extraction: Perform an `INNER JOIN` between the `equipment` table and the `issue_book` (transaction) table.

2. Aggregation: Group the results by `equipment_id`.

3. Count Calculation: Use the `COUNT()` function to determine how many times each item has been requested.

4. Ranking: Sort the grouped results in descending order based on the request count (`borrow_count`).

5. Selection: Limit the results to the top 3 items.

6. Output: Display these items in a "Recommended Equipments" gallery on the user dashboard.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

In this phase, theoretical design is turned into practice i.e. working system. The most crucial stage in achieving a new successful system and in giving confidence on the new system for the users that it will work efficiently and effectively.

The system can be implemented only after thorough testing is done and if it is found to work according to the specification.

4.1.1 Tools Used

Following are the tools and framework used for the accomplishment of this project:

Front END Tools

Front end tools involve the interface user interacts on the screen. The front-end tools used are as follows:

- **HTML**

  In Sports Equipment Rental System, HTML has been used for structuring the content of the web application. It is used to add headings, paragraphs, links, and images to a webpage.

- **CSS**

  CSS is used for styling and designing visual elements of the application. Different CSS are used such as inline CSS, internal CSS, and external CSS to design this system. It is used for defining the style for web pages. By using CSS, we can control the text color, font style, the spacing between paragraphs, sizing of columns, layout designs, and many more.

- **JavaScript**

  JS (JavaScript) is a lightweight, interpreted programming language designed for creating network-centric applications. It is complimentary to and integrated with Java. JavaScript is very easy to implement because it is integrated with HTML. It is open and cross-platform.

- **Bootstrap**

  It is one of the most popular HTML, CSS and JavaScript Framework for developing built responsive, mobile-first websites. The primary purpose of adding it to a web project is to apply Bootstrap's choices of color, size, font and layout to that project.

Back End Tools

Backend tools are the set of technologies, frameworks, programming languages, and software utilities used to create the "under-the-hood" portion of a website, application, or system.

- **PHP**

  PHP is used for the backend purpose and for making dynamic web pages. It is used for server-side scripting purpose to add connectivity to the database and also used to encrypt the data, validate the user data, confirm user to go to certain pages, recommend equipment to the user, fine calculation and so on. It also includes add, update and delete the data from the database.

Database Tools

Database tools are software applications and utilities used to design, manage, administer, and maintain databases, enabling users to interact with data efficiently.

- **MYSQL**

  MYSQL is used for storing all the information required to the database in Sports Equipment Rental System. It is used for performing CRUD operation which includes creating, reading, updating and deleting of data from the database as requested by the user.

Documentation Tools

Documentation tools are software applications used to create, manage, organize, and share information, such as technical manuals, project guides, and internal processes.

- **Draw.io**

  Draw.io is used to generate diagrams for the system analysis and design of Sports Equipment Rental System like Entity-Relationship Diagrams, Data Flow Diagrams and so on. Diagrams were created using these tools in order to save time since all components are available with drag and drop functions.

- **Figma**

  Figma is an open-source free web-based UI tool to create, collaborate, prototype, and handoff. It has been in use to design the interface of the project SERS as a UI/UX Designing tool.

IDE and Code Editor

An IDE (Integrated Development Environment) is a comprehensive software suite that combines all the essential tools for software development into a single interface. A Code Editor is a lightweight tool primarily focused on the task of writing and editing text-based source code.

- **Visual Studio Code:**
  Visual Studio Code served as the primary tool for developing and writing the source code of the System. This powerful code editor provided an interactive development environment, enhancing productivity.

4.1.2 Implementation Details of Modules

The SERS system is built using a modular approach where each section of the application handles a specific part of the rental process. The implementation details for these modules are as follows:

- **Authentication and User Management Module**:

This module is responsible for controlling access to the system for both students and administrators. It is implemented using PHP session variables that track the user's identity as they move from page to page. During the login process, the system executes a SQL query to check the provided credentials against the "user" or "admin" tables. If a match is found, the system initializes the session and redirects the user to their respective dashboard, ensuring that only authorized individuals can request equipment or manage the inventory.

- **Equipment Inventory Module:**

The inventory module handles the storage and display of all sports gear available for rent. It is implemented with a focus on real-time data accuracy, where the administrator can add new items, update existing details, or remove damaged equipment through a secure interface. The system uses PHP loops to fetch these records from the database and display them in a structured table for users. This implementation ensures that users can see the exact quantity available and the current condition of the equipment before making a request.

- **Request and Approval Transaction Module:**

This module manages the core borrowing process, acting as a bridge between the user's needs and the administrator's oversight. When a user requests an item, the implementation

creates a new entry in the transaction table with a 'Pending' status. The administrator can then review these requests and, upon approval, the system simultaneously updates the transaction status and decrements the available quantity in the inventory. This dual-update logic is critical to maintaining data integrity across the entire platform.

- **Fine Calculation and Tracking Module:**

To ensure the timely return of borrowed items, this module automates the monitoring of return deadlines. The implementation compares the current system date with the expected return date stored in the transaction records. If an item is overdue, the system calculates a fine based on the number of days delayed and updates the fine table accordingly. This automated approach reduces manual paperwork for the administrator and provides users with a transparent view of any penalties they have incurred.

- **Popularity-Based Recommendation Module:**

This module is implemented to improve the user experience by suggesting the most relevant equipment to users. It uses a popularity-based algorithm that performs an aggregation on the transaction history to identify which items are borrowed most frequently. The top three items are then fetched and displayed on the user dashboard. This implementation helps in optimizing equipment usage by highlighting high-demand gear to all users.

4.2 Testing

System testing is done by giving different training and testing datasets. This test is done to evaluate whether the system is providing accurate summary or not. The series of testing conducted are as follows:

4.2.1 Test Cases for Unit Testing

In unit testing, we designed the entire system in modularized pattern and each module is tested. Until we get the accurate output from the individual module, we work on the same module. The input forms are tested so that they do not accept invalid input. The unit testing that has been carried out in form of user login and user registration are given in the following tables:

Table 1 Test Case for User Registration

SN	Test Case	Test Description	Test Inputs	Expected Outcome	Observed Outcome	Result
1	User register case-i	User registers with empty fields	None	Error message for empty fields will appear	Please fill out this field message	PASS
2	User register case – ii	User registers with wrong input formats	First Name: Diwas Last Name: Lamichhane Username: diwasey Email:diwaslamichhane@gmail.com Mobile Number: 980000 Password: diwas Mobile Number: 9800000000 Password: diwasey	Alert box with message Denoting error in length of mobile number The password length error message will appear Username pre-existing error	Mobile number must be 10 digits Password must be at least 6 characters Username already exists.	PASS
3	User register case – iii	User correctly inputs the fields	First Name: Diwas Last Name: Lamichhane Username: diwasdai Email:diwaslamichhane@gmail.com Mobile Number: 9800000000 Password: diwasey	Registration Successful		PASS

Table 2 Test Case for User Login

SN	Test Case	Test Description	Test Inputs	Expected Outcome	Observed Outcome	Result
1	User login case - i	User tries logging with empty fields	None	Please fill out this field message	Please fill out this field message	PASS
2	User login case - ii	User enters wrong password	Username: diwasey Password: diwas	The user username and password doesn't match	The system displayed: "The user username and password doesn't match"	PASS
3	User login case – iii	User enter correct credentials	Username: diwasdai Password: diwasey	Redirect to User dashboard i.e. home page	The user was successfully redirected to the dashboard (home page)	PASS

Table 2 presents the test cases conducted for the User Login module of the Sports Equipment Rental System (SERS). The testing was carried out across three different login scenarios to ensure the authentication system functions correctly under varying conditions. All three test cases produced the expected outcomes, and each received a PASS result, indicating that the login module of SERS is fully functional, secure, and user-friendly.

4.2.2 Test Cases for System Testing

Table 3 Test Cases for System Testing

SN	Test Case	Description	Test Inputs	Expected Result	Observed Outcome	Result
1	System Launch	Accessing the home page via a local server.	URL: local host/sers/	The landing page loads successfully with the navigation bar.	Landing page loaded successfully with all components.	PASS
2	User Authentication	Entering valid username & password.	Username: diwasdai, Password: diwasey	System validates and routes to user dashboard.	Credentials verified and dashboard accessed.	PASS
3	Equipment Browsing	Navigating to the catalog to view sports gear.	Click "View Equipment"	Displays a list of equipment with category, quantity, and price.	Full equipment list displayed with correct data.	PASS
4	Search Functionality	Using the search bar to find a specific item.	Input: "Cricket Bat"	The system filters the list and shows only matching items.	Search returned only relevant "Cricket Bat" results.	PASS
5	Borrow Request	Clicking the "Request" button for equipment.	Click “Request”	A request is generated and added to 'Pending'	Request recorded in database under 'Pending' status.	PASS
6	Admin Approval	Admin reviews and clicks "Approve".	Admin Action: Approve	Status changes to 'Yes' and equipment quantity decreases by 1.	Status updated and stock count correctly reduced.	PASS

SN	Test Case	Description	Test Inputs	Expected Result	Observed Outcome	Result
7	Recommendation	Accessing dashboard after transactions.	User Login	System displays 'Top 3' borrowed items.	Dashboard showed personalized equipment suggestions.	PASS
8	Fine Calculation	System date exceeds return date.	Date > Return Date	System calculates daily penalty and updates table.	Fine accurately applied based on overdue days.	PASS
9	Security Access	Attempting direct access to admin files.	URL: admin/index.php	Redirects to login page (session detection).	Unauthorized access blocked; redirected to login.	PASS
10	Logout Procedure	Clicking the logout button.	Click "Logout"	Session destroyed; redirected to home page.	User session cleared and returned to home page.	PASS

The system testing phase for the Scientific Equipment Rental System (SERS) was conducted to ensure that all integrated modules function cohesively as per the requirement specifications. The testing covered critical functional areas including User Authentication, Inventory Management, Request Workflows, and Automated Fine Calculations.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

The development of the Sports Equipment Rental System (SERS) has been successfully completed. This project was designed to replace the old, manual way of tracking sports and scientific gear with a modern, automated web application. By using PHP and MySQL, I have built a system where students can easily browse and request equipment, while administrators can manage the inventory and track rentals in one place.

In conclusion, the system meets all its initial goals. It provides a user-friendly interface, ensures data security through a login system, and automates difficult tasks like fine calculation and equipment recommendations. SERS makes the rental process faster, more transparent, and much more efficient for everyone involved.

5.2 Lesson Learnt / Outcome

Working on this project has been a great learning experience. I have gained practical skills in several areas of web development:

- **Full-Stack Development:** I learned how to connect a frontend (HTML/CSS) with a backend (PHP) and a database (MySQL) to create a working application.
- **Database Management:** I understood how to design a database schema and use SQL queries to handle complex data like transactions and equipment counts.
- **Problem Solving:** I faced and solved real-world coding challenges, such as implementing logic for calculating fines and creating a recommendation system based on user behavior.
- **System Design:** I learned the importance of planning a system using DFDs and ER diagrams before actually starting the coding phase.

5.3 Future Recommendations

While SERS is fully functional, there are several ways it can be improved in the future to make it even more powerful:

- **Online Payment Integration:** Adding a payment gateway (like Khalti or Esewa) would allow users to pay their fines directly through the website.
- **SMS Notifications:** The system could be updated to send automatic text message reminders to users when their return date is approaching.
- **Barcode Scanning:** Implementing barcode or QR code scanning would make the issuing and returning of equipment much faster for the administrator.
- **Mobile Application:** Developing a dedicated Android or iOS app would make it easier for students to browse the equipment catalog on the go.
- **Multi-Language Support:** Adding support for local languages could make the system more accessible to a wider range of users.

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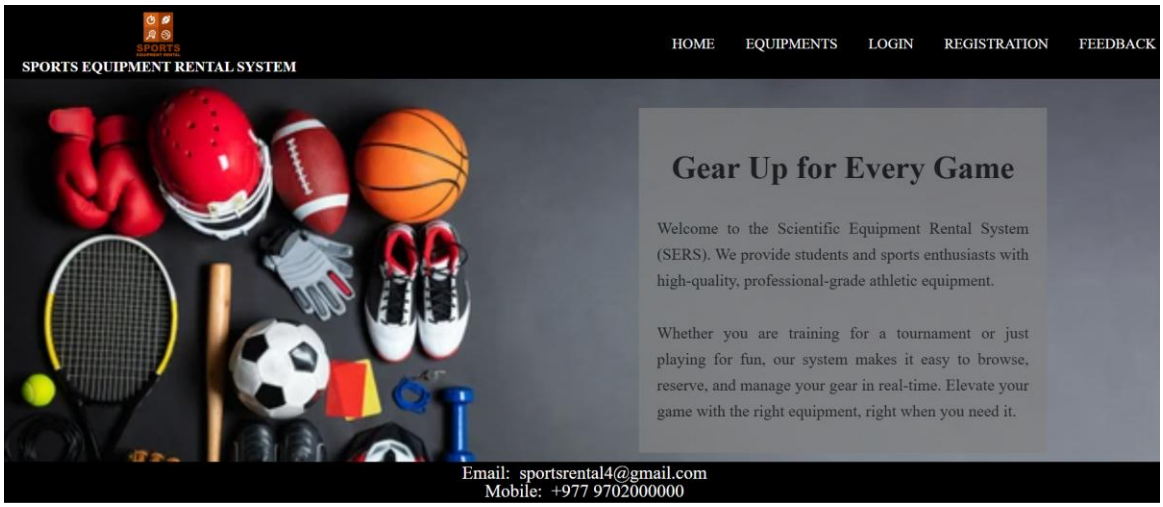
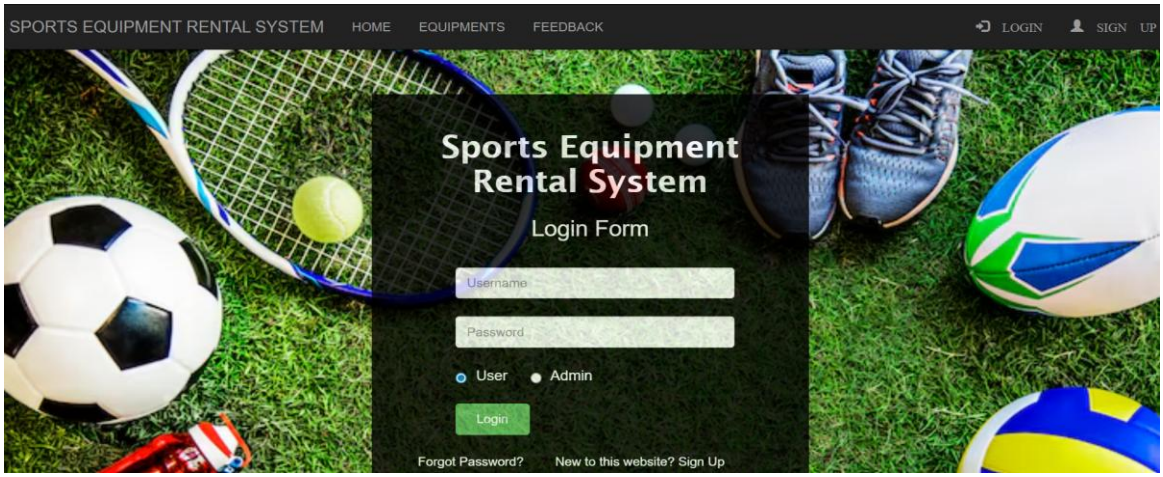
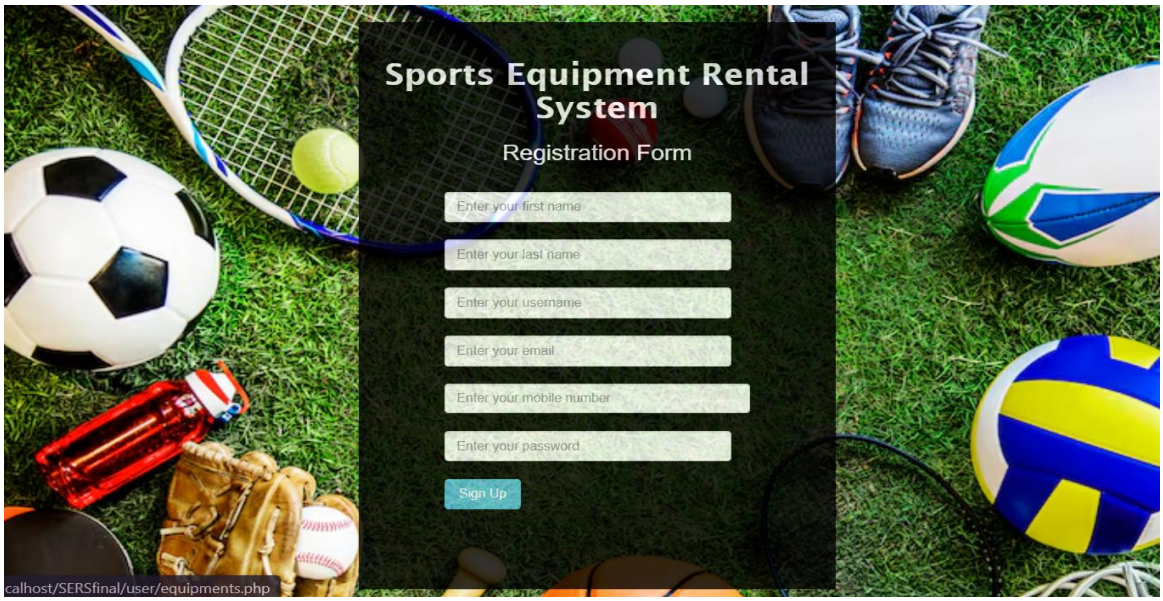
APPENDICES



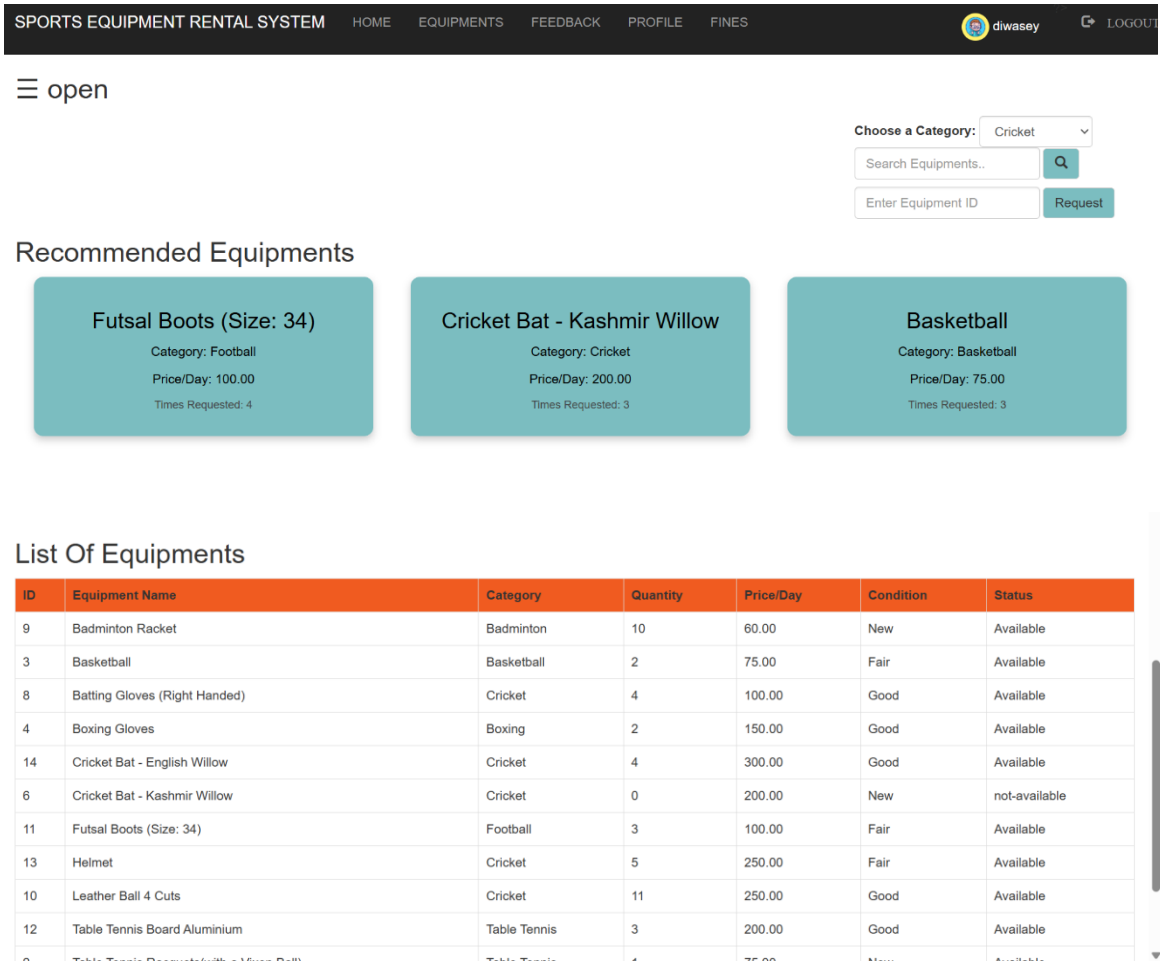
Home Page



Login Page

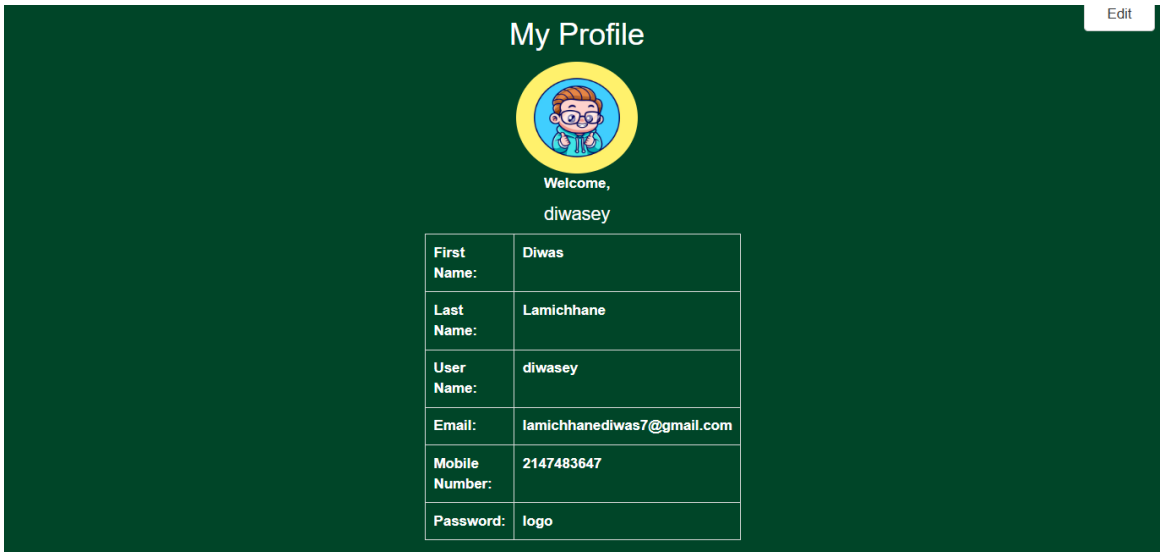


User Registration Page

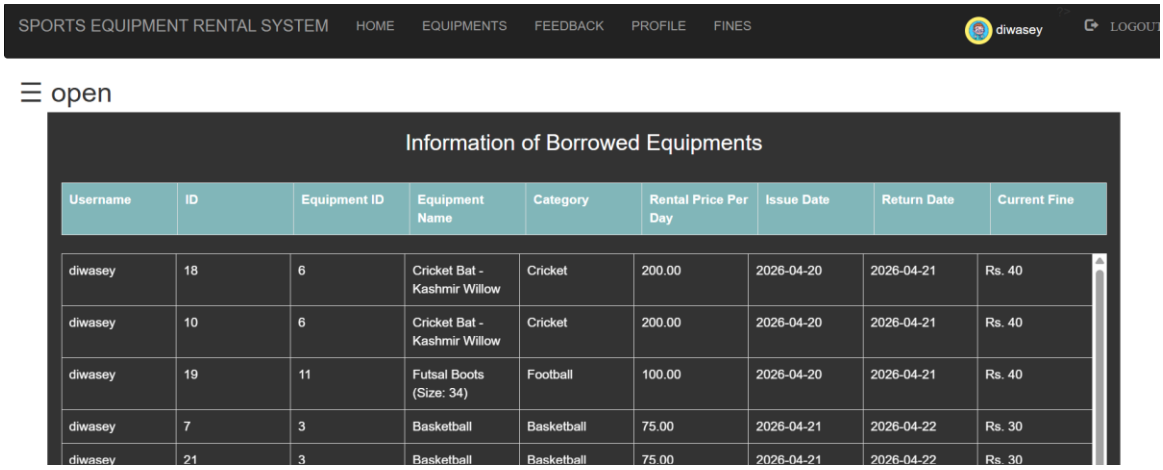


ID	Equipment Name	Category	Quantity	Price/Day	Condition	Status
9	Badminton Racket	Badminton	10	60.00	New	Available
3	Basketball	Basketball	2	75.00	Fair	Available
8	Batting Gloves (Right Handed)	Cricket	4	100.00	Good	Available
4	Boxing Gloves	Boxing	2	150.00	Good	Available
14	Cricket Bat - English Willow	Cricket	4	300.00	Good	Available
6	Cricket Bat - Kashmir Willow	Cricket	0	200.00	New	not-available
11	Futsal Boots (Size: 34)	Football	3	100.00	Fair	Available
13	Helmet	Cricket	5	250.00	Fair	Available
10	Leather Ball 4 Cuts	Cricket	11	250.00	Good	Available
12	Table Tennis Board Aluminium	Table Tennis	3	200.00	Good	Available

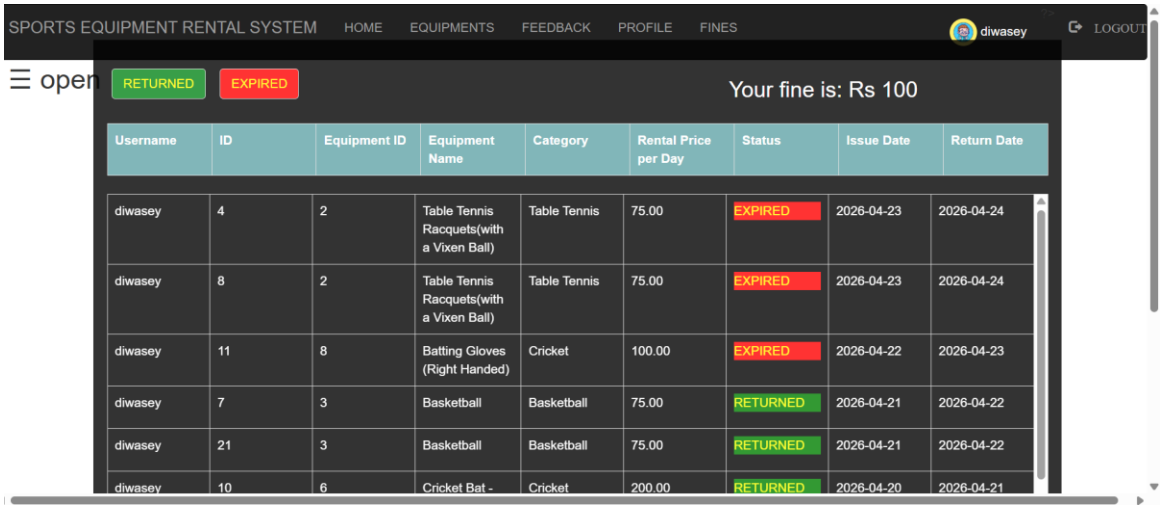
Equipment Page (User Side)



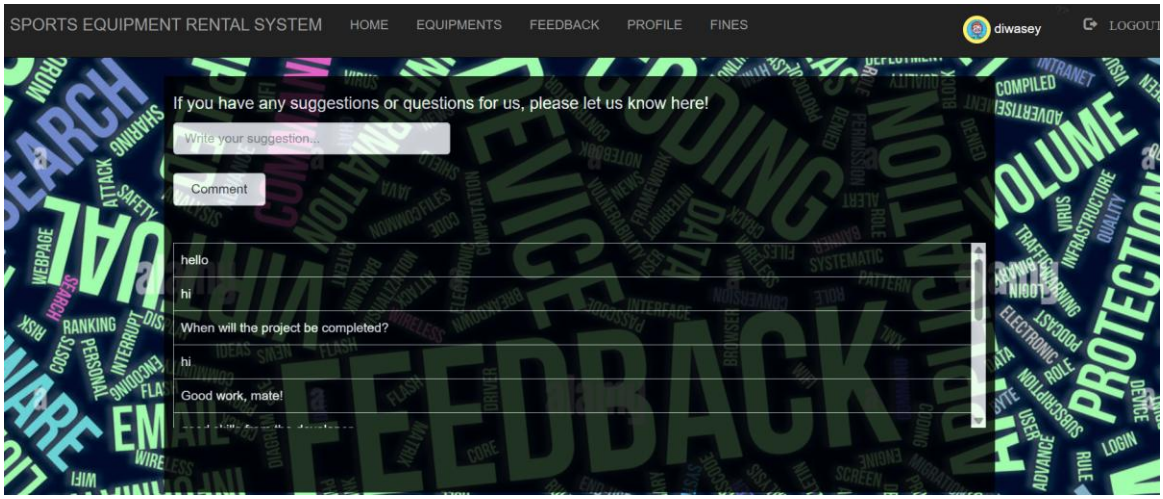
User Profile Page



User Issue Information Page



User Expired List



User Feedback Page

SPORTS EQUIPMENT RENTAL SYSTEM HOME EQUIPMENTS FEEDBACK USER-INFORMATION FINES daitya LOGOUT

Search Username...

List Of Users

ID	First Name	Last Name	Username	Email	Mobile Number
9	abc	abc	def	abc@gmail.com	2147483647
3	Diwas	Lamichhane	diwasey	lamichhanediwas7@gmail.com	2147483647
11	Diwas	Lamichhane	diwasdai	diwaslamichhane@gmail.coma	2147483647
5	Elliot	Alderson	elliot	mrrobot@gmail.com	2147483647
10	Patrick	Bateman	patrick	bateman@gmail.com	2147483647

User Information Page (Admin Side)

SPORTS EQUIPMENT RENTAL SYSTEM HOME EQUIPMENTS FEEDBACK USER-INFORMATION ADMINS FINES LOGOUT daitya

open

User username..

List Of Users

Username	Equipment id	Returned	Days	Fines in Rs	Status	Action
patrick	10	2026-04-23	0	0	paid	
patrick	9	2026-04-23	1	0	not paid	Mark Paid
patrick	11	2026-04-23	1	0	paid	
patrick	11	2026-04-23	1	1	paid	
diwasey	3	2026-04-23	2	1	paid	
diwasey	11	2026-04-23	2	20	paid	
diwasey	6	2026-04-23	2	20	paid	
diwasey	6	2026-04-23	2	20	paid	

Fines Page (Admin Side)

SPORTS EQUIPMENT RENTAL SYSTEM HOME EQUIPMENTS FEEDBACK USER-INFORMATION ADMINS FINES LOGOUT daitya

Add New Admin

First Name

Last Name

Username

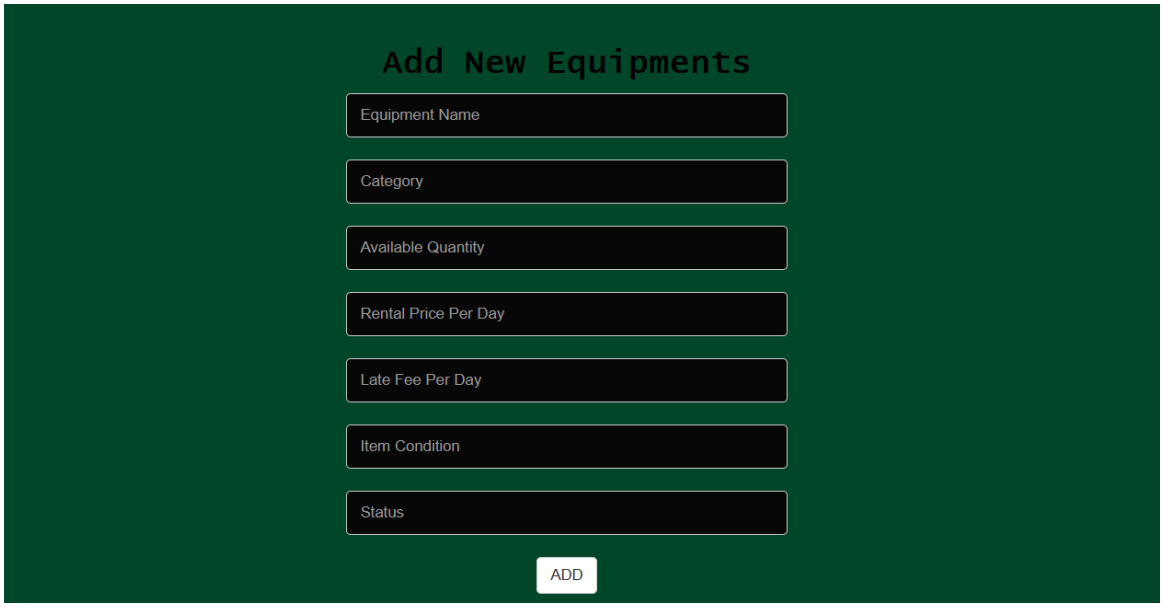
Email

Mobile Number

Current Administrators

ID	Full Name	Username	Email	Contact
6	Nishab Rupakheti	jptho	ksjdhcgf@gmail.coma	0909
5	Nishab Rupakheti	nishab	rupakhetinishab@gmail.com	9702038475
4	abcdef ghijkl	abcde	abcde@gmail.com	0909090909
3	Test Bahadur	testadmin	admin@gmail.com	9899
2	aaa aaa	aaa	aaa	111
1	Diwas Lamichhane	daitya	cristianodeewosh@gmail.com	9800000000

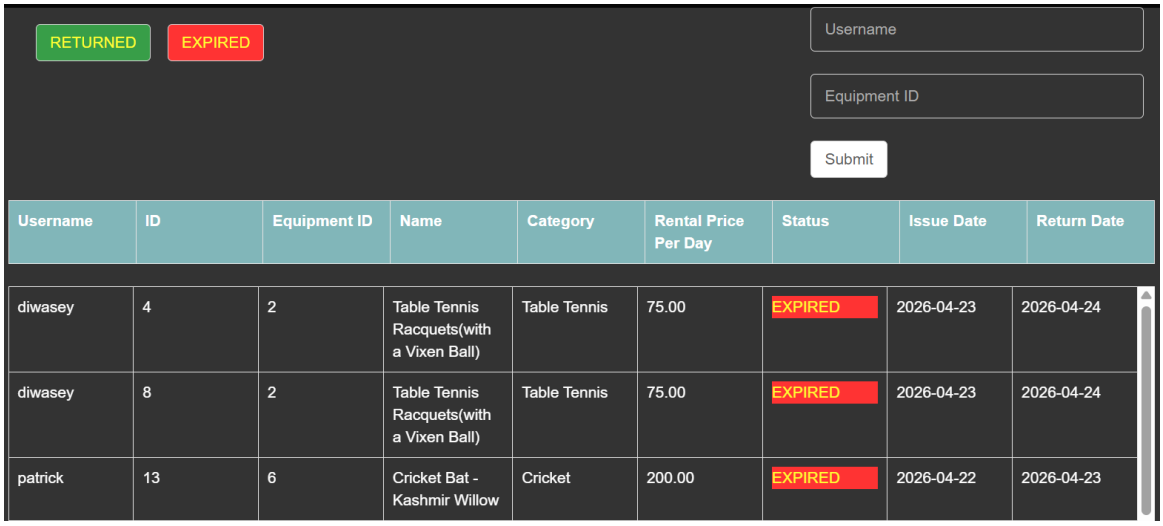
Admins Page



Add Equipments Page

Username	ID	Equipment ID	Equipment Name	Category	Rental Price Per Day	Status
diwasey	23	14	Cricket Bat - English Willow	Cricket	300.00	Available

Equipment Request Page (Admin Side)



Username	ID	Equipment ID	Name	Category	Rental Price Per Day	Status	Issue Date	Return Date
diwasey	4	2	Table Tennis Racquets(with a Vixen Ball)	Table Tennis	75.00	EXPIRED	2026-04-23	2026-04-24
diwasey	8	2	Table Tennis Racquets(with a Vixen Ball)	Table Tennis	75.00	EXPIRED	2026-04-23	2026-04-24
patrick	13	6	Cricket Bat - Kashmir Willow	Cricket	200.00	EXPIRED	2026-04-22	2026-04-23

Issue Information Page (Admin Side)